

National Coverage Determination (NCD)

Beta Amyloid Positron Tomography in Dementia and Neurodegenerative Disease

220.6.20

Links in PDF documents are not guaranteed to work. To follow a web link, please use the MCD Website.

Tracking Information

Publication Number

100-3

Manual Section Number

220.6.20

Manual Section Title

Beta Amyloid Positron Tomography in Dementia and Neurodegenerative Disease

Version Number

1

Effective Date of this Version

09/27/2013

Ending Effective Date of this Version

10/13/2023

Implementation Date

07/07/2014

Description Information

Benefit Category

Diagnostic Tests (other)

Please Note: This may not be an exhaustive list of all applicable Medicare benefit categories for this item or service.

Item/Service Description

A. The Centers for Medicare & Medicaid Services (CMS) has determined that the evidence is insufficient to conclude that the use of positron emission tomography (PET) beta amyloid (also referred to as amyloid-beta (A β)) imaging is reasonable and necessary for the diagnosis or treatment of illness or injury or to improve the functioning of a malformed body member for Medicare beneficiaries with dementia or neurodegenerative disease, and thus PET A β imaging is not covered under §1862(a)(1)(A) of the Social Security Act ("the Act").

Indications and Limitations of Coverage

B. However, there is sufficient evidence that the use of PET A β imaging is promising in two scenarios: (1) to exclude Alzheimer's disease (AD) in narrowly defined and clinically difficult differential diagnoses, such as AD versus frontotemporal dementia (FTD); and (2) to enrich clinical trials seeking better treatments or prevention strategies for AD, by allowing for selection of patients on the basis of biological as well as clinical and epidemiological factors.

Therefore, we will cover one PET A β scan per patient through coverage with evidence development (CED), under §1862(a)(1)(E) of the Act, in clinical studies that meet the criteria in each of the paragraphs below.

Clinical study objectives must be to (1) develop better treatments or prevention strategies for AD, or, as a strategy to identify subpopulations at risk for developing AD, or (2) resolve clinically difficult differential diagnoses (e.g., frontotemporal dementia (FTD) versus AD) where the use of PET A imaging appears to improve health outcomes. These may include short term outcomes related to changes in management as well as longer term dementia outcomes.

Clinical studies must be approved by CMS, involve subjects from appropriate populations, and be comparative and longitudinal. Where appropriate, studies should be prospective, randomized, and use postmortem diagnosis as the endpoint. Radiopharmaceuticals used in the PET A β scans must be FDA approved. Approved studies must address one or more aspects of the following questions. For Medicare beneficiaries with cognitive impairment suspicious for AD, or who may be at risk for developing AD:

1. Do the results of PET A β imaging lead to improved health outcomes? Meaningful health outcomes of interest include: avoidance of futile treatment or tests; improving, or slowing the decline of, quality of life; and survival.
2. Are there specific subpopulations, patient characteristics or differential diagnoses that are predictive of improved health outcomes in patients whose management is guided by the PET A β imaging?

3. Does using PET A β imaging in guiding patient management, to enrich clinical trials seeking better treatments or prevention strategies for AD, by selecting patients on the basis of biological as well as clinical and epidemiological factors, lead to improved health outcomes?

Any clinical study undertaken pursuant to this national coverage determination (NCD) must adhere to the timeframe designated in the approved clinical study protocol. Any approved clinical study must also adhere to the following standards of scientific integrity and relevance to the Medicare population.

- a. The principal purpose of the research study is to test whether a particular intervention potentially improves the participants' health outcomes.
- b. The research study is well supported by available scientific and medical information or it is intended to clarify or establish the health outcomes of interventions already in common clinical use.
- c. The research study does not unjustifiably duplicate existing studies.
- d. The research study design is appropriate to answer the research question being asked in the study.
- e. The research study is sponsored by an organization or individual capable of executing the proposed study successfully.
- f. The research study is in compliance with all applicable Federal regulations concerning the protection of human subjects found at 45 CFR Part 46. If a study is regulated by the Food and Drug Administration (FDA), it must be in compliance with 21 CFR parts 50 and 56.
- g. All aspects of the research study are conducted according to appropriate standards of scientific integrity (see <http://www.icmje.org> ) .
- h. The research study has a written protocol that clearly addresses, or incorporates by reference, the standards listed here as Medicare requirements.
- i. The clinical research study is not designed to exclusively test toxicity or disease pathophysiology in healthy individuals. Trials of all medical technologies measuring therapeutic outcomes as one of the objectives meet this standard only if the disease or condition being studied is life threatening as defined in 21 CFR §312.81(a) and the patient has no other viable treatment options.
- j. The clinical research study is registered on the ClinicalTrials.gov website by the principal sponsor/investigator prior to the enrollment of the first study subject.

- k. The research study protocol specifies the method and timing of public release of all pre-specified outcomes to be measured including release of outcomes if outcomes are negative or the study is terminated early. The results must be made public within 24 months of the end of data collection. If a report is planned to be published in a peer reviewed journal, then that initial release may be an abstract that meets the requirements of the International Committee of Medical Journal Editors (<http://www.icmje.org>). However a full report of the outcomes must be made public no later than three (3) years after the end of data collection.
- l. The research study protocol must explicitly discuss subpopulations affected by the treatment under investigation, particularly traditionally underrepresented groups in clinical studies, how the inclusion and exclusion criteria effect enrollment of these populations, and a plan for the retention and reporting of said populations on the trial. If the inclusion and exclusion criteria are expected to have a negative effect on the recruitment or retention of underrepresented populations, the protocol must discuss why these criteria are necessary.
- m. The research study protocol explicitly discusses how the results are or are not expected to be generalizable to the Medicare population to infer whether Medicare patients may benefit from the intervention. Separate discussions in the protocol may be necessary for populations eligible for Medicare due to age, disability or Medicaid eligibility.

Consistent with §1142 of the Act, the Agency for Healthcare Research and Quality (AHRQ) supports clinical research studies that CMS determines meet the above-listed standards and address the above-listed research questions.

All other uses are noncovered.

Cross Reference

[CED page](#) 

Claims Processing Instructions

[TN 2871 \(Medicare Claims Processing\)](#) 

[TN 2915 \(Medicare Claims Processing\)](#) 

[TN 12364 \(Medicare Claims Processing\)](#) 

Transmittal Information

Transmittal Number

164

Coverage Transmittal Link

<https://www.cms.gov/Regulations-and-Guidance/Guidance/Transmittals/Downloads/R164NCD.pdf> 

Revision History

10/2022 - Transmittal 11546, dated August 4, 2022, is being rescinded and replaced by Transmittal 11636, dated, October 5, 2022, to remove ICD-10 dx codes added in error to NCD 150.3, business requirement 12842.4, and restore ICD-10 dx C91.92 removed in error to NCD 110.23, business requirement 12842.3. All other information remains the same. ([TN 11636](#) ) (CR12842)

08/2022 - The purpose of this Change Request (CR) is to provide a maintenance update of ICD-10 conversions and other coding updates specific to NCDs. ([TN 11546](#) ) (CR12842)

07/2017 - This Change Request (CR) constitutes a maintenance update of International Code of Diseases, Tenth Revision (ICD-10) conversions and other coding updates specific to National Coverage Determinations (NCDs). These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received.

Previous NCD coding changes appear in ICD-10 quarterly updates that can be found at: <https://www.cms.gov/Medicare/Coverage/CoverageGenInfo/ICD10.html> , along with other CRs implementing new policy NCDs. Edits to ICD-10 and other coding updates specific to NCDs will be included in subsequent quarterly releases and individual CRs as appropriate. No policy-related changes are included with the ICD-10 quarterly updates. Any policy-related changes to NCDs continue to be implemented via the current, long-standing NCD process. ([TN 1875](#) ) (CR10184)

11/2016 - This change request (CR) is the 9th maintenance update of ICD-10 conversions and other coding updates specific to national coverage determinations (NCDs). The majority of the NCDs included are a result of feedback received from previous ICD-10 NCD CR7818, CR8109, CR8197, CR8691, CR9087, CR9252, CR9540, and CR9631. Some are the result of revisions required to other NCD-related CRs released separately.

Edits to ICD-10 and other coding updates specific to NCDs will be included in subsequent, quarterly releases as needed. No policy-related changes are included with these updates. Any policy-related changes to NCDs continue to be implemented via the current, long-standing NCD process. ([TN 1753](#) ) (CR9751)

08/2016 - This change request (CR) is the 9th maintenance update of ICD-10 conversions and other coding updates specific to national coverage determinations (NCDs). The majority of the NCDs included are a result of feedback received from previous ICD-10 NCD CR7818, CR8109, CR8197, CR8691, CR9087, CR9252, CR9540, and CR9631. Some are the result of revisions required to other NCD-related CRs released separately.

Edits to ICD-10 and other coding updates specific to NCDs will be included in subsequent, quarterly releases as needed. No policy-related changes are included with these updates. Any policy-related changes to NCDs continue to be implemented via the current, long-standing NCD process. ([TN 1708](#)) (CR9751)

03/2014 - Transmittal 160, dated February 6, 2014 is being rescinded and replaced by Transmittal 164, dated March 27, 2014 to update the reference to the correct CPM chapter 13 in the first business requirement. All other information remains the same. Effective date: 09/27/2013. Implementation date: 07/07/2014. ([TN 164](#)) (CR8526)

02/2014 - The purpose of this Change Request (CR) is that effective for claims with dates of service on or after, September 27, 2013, Medicare will only allow coverage for PET Aβ imaging (one PET Aβ scan per patient) through coverage with evidence development (CED) to: (1) develop better treatments or prevention strategies for AD, or, as a strategy to identify subpopulations at risk for developing AD, or (2) resolve clinically difficult differential diagnoses (e.g., frontotemporal dementia (FTD) versus AD) where the use of PET Aβ imaging appears to improve health outcomes, when the patient is enrolled in an approved clinical study under CED. Effective date: 09/27/2013. Implementation date: 07/07/2014. ([TN 160](#)) (CR8526)

National Coverage Analyses (NCAs)

This NCD has been or is currently being reviewed under the National Coverage Determination process. The following are existing associations with NCAs, from the National Coverage Analyses database.

- [Original Consideration for Beta Amyloid Positron Emission Tomography in Dementia and Neurodegenerative Disease \(CAG-00431N\)](#)
- [First reconsideration for Beta Amyloid Positron Emission Tomography in Dementia and Neurodegenerative Disease \(CAG-00431R\)](#)

Additional Information

Other Versions

Title	Version	Effective Between	
Beta Amyloid Positron Tomography in Dementia and Neurodegenerative Disease (RETIRED)	2	10/13/2023 - N/A	View
Beta Amyloid Positron Tomography in Dementia and Neurodegenerative Disease	1	09/27/2013 - 10/13/2023	You are here